

Adaline Powell

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EDUCATION

University of Wisconsin–Milwaukee December 2026

Master of Science, Data Science and Applied AI

University of Wisconsin–Milwaukee May 2025

Master of Arts, Political Science

University of Wisconsin–Milwaukee. May 2023

Bachelor of Arts, Political Science (Pre-Law) | Bachelor of Arts, Philosophy

WORK EXPERIENCE

Research Assistant

September – December 2025

Argosy Foundation · Milwaukee, WI · How Are Foundations Navigating Uncertainty?

- Analyzed philanthropic foundation public statements to identify responses to economic instability and government budget cuts.
- Conducted NLP, text analysis, and topic modeling in Python and R to extract themes; identified 2 broad themes and 5 subthemes.
- Presented results through an infographic and stakeholder presentation to support strategic discussion.

Research Assistant

June – August 2023

Argosy Foundation · Milwaukee, WI · Nonprofit Financial Practices Analysis

- Evaluated how nonprofit savings levels affect grant eligibility using a sample from Argosy's grant portfolio.
- Modeled funding scenarios to estimate qualification rates under different criteria; visualized results with summary tables and charts.
- Developed and presented recommendations for savings ratio–based grant-making guidelines to support internal decision-making.

PROJECTS

Dream Machine Live Data Dashboard — Discovery World Hackathon

February 2026

UWM Connected Systems Institute × Northwestern Mutual Data Science Institute

- Served as Data Engineer at a competitive hackathon with Rockwell Automation and Microsoft; built a real-time IoT ELT pipeline on Microsoft Fabric streaming PLC sensor data through Azure IoT Hub into a KQL Medallion architecture.
- Wrote scheduled PySpark notebooks using window functions and lag operations to detect time-series state transitions — identifying cycle completions and laser cut times from raw sensor data.
- Decoded guest order data from packed hex byte strings (e.g. b'Turtle.dat\x00\x00ALICE\x0042') using custom KQL parsing functions to extract design, creator name, and order number into usable fields.
- Decoded machine fault states from 32-bit integers bit-by-bit against a fault lookup table to surface real-time system health in plain language on a live Power BI exhibit display.

Milwaukee Birth Trends Dashboard January 2026

- Built an interactive dashboard analyzing birth trends in Milwaukee using Wisconsin DHS WISH data; used Databricks and SQL to clean, transform, and aggregate data across multiple tables.
- Developed visualizations including line graphs, bar plots, and pie charts to examine longitudinal trends and demographic differences.

YouTube Gaming Engagement Analysis December 2026

- Analyzed engagement patterns across trending YouTube gaming videos using YouTube API data; performed data cleaning, feature engineering, and EDA in R and Python.

- Engineered content features including title characteristics via regex, clickbait signals via spaCy NLP part-of-speech tagging, and thumbnail analysis using OpenCV face detection and DeepFace emotion recognition.
- Built and compared regression models, logistic models, ordinal/multinomial models, K-means clustering, and neural networks; evaluated using cross-validation and hold-out testing.
- Translated modeling results into practical content strategy recommendations for improving video performance.

Political Science Survey Research and Experimental Design

May 2025

- Designed an original survey in Qualtrics distributed via Lucid, collecting responses from 1,012 participants with an embedded experimental design manipulating question wording to examine partisan strength.
- Analyzed survey data in R incorporating demographic, behavioral, and attitudinal measures; presented findings as part of a funded graduate research seminar.

TECHNICAL SKILLS

Data Engineering & Platforms: ELT Pipelines, KQL (Kusto Query Language), PySpark, Azure IoT Hub, Microsoft Fabric (Eventhouse, Eventstreams), Databricks, SQL, Power BI, Streaming Data, Real-Time Analytics, ArcGIS

Data Science & Modeling: Machine Learning, AI Integration, NLP, Regression Analysis, Predictive Modeling, Feature Engineering, Cross-Validation, K-Means, Random Forests, XGBoost, Neural Networks, Data Visualization

Programming & Tools: Python, R, MySQLWorkbench, SPSS, Qualtrics

Certificates: Quantitative Social Data Analysis

EXTRACURRICULARS

Northwestern Mutual Data Science Institute

Women's Track and Field